

Assessment Support Report Guidance

Abstract

- 5 %
- Max 150 words
- Recommendation to write this last!



Purpose: Provide a self-contained summary of the project.



You should include:

Why the study was done (e.g., importance of vertical jump analysis and comparing jump height methods)

How the data was collected (briefly mention force and marker data from CMJ and SQJ)

Key results (e.g., which method most closely estimated jump height)

Main conclusion (e.g., best estimation method and implications)

Introduction

- 10 %



Tips:

- Use min 4–6 good quality sources (journal articles/textbooks).
- Include recent literature where possible.
- Use the introduction to justify why this comparison matters.

Set the context and justify the study with relevant literature.

Should cover:

- Why vertical jump performance is important in biomechanics/biomedical applications.
- Common ways to measure vertical jump height.
- Brief overview of the three estimation methods:
 - Kinematic
 - Projectile motion
 - Conservation of energy
- Differences between **CMJ** and **SJ** — theoretical background and prior research.
- Importance of using centre of mass (CoM) data as a control.

Methods

- 30 %



Tips:

- Write in the past tense
- Don't include code or screenshots.
- Use **equations & figures** to explain complex concepts.
- Justify methods using references

Describe how the analysis was performed in enough detail for replication.

Should include the following subsections:

- Participants
- Instrumentation
- Data Collection
- Data Analysis

Methods

- Participants

- Number of participants
- Basic demographics (age, sex, height, mass)
- Any inclusion or exclusion criteria
- Ethical approval line such as
Data were collected as part of a scheduled teaching activity.
Ethical approval was granted by the institution, and informed consent was obtained

Methods

- Instrumentation
 - 12 Vicon Vero v2.2 cameras ((Vicon Motion Systems, Oxford, UK) sampling at 100 Hz
 - 1 AMTI OR/6-5-1000 force plate (Advanced Mechanical Technology Inc., Watertown, MA, USA) sampling at 1000 Hz

Methods

- Data collection

Tips:

- Use references to justify your protocol

- Marker set – an image would be beneficial here
- How were markers attached
- Static trial and dynamic trials
- Types of jumps performed
- Trial conditions (e.g., shoes/barefoot, hands on hips)
- Any calibration procedures – brief
- Warm up & familiarisation trials

Methods

- Data Analysis



Tips:

- Use references to support and justify your data analysis approach

- Use of Vicon Nexus software
- Labelling & Gap filling (rigid body, kinematic and pattern fills)
- Filtering – conducted prior to export of data
 - Trajectory data: 4th order low pass Butterworth 6 Hz cut off
 - Force data: 4th order low pass Butterworth 12 Hz cut off
- Exportation of data and use of custom code in Matlab (R2023a, Mathworks)
- Resampling of force data from 1000 Hz to 100 Hz
- Outline how each method estimated jump height
 - Gold standard method – vertical displacement of whole body CoM
 - Kinematic - vertical displacement from Newton's 3rd law and integration of accel data
 - Projectile motion - using take-off velocity
 - Conservation of energy converting KE to PE)
 - Use of virtual markers to estimate CoM, or assumption using RASI/LASI/RPSI/LPSI midpoint.
- Use of Dempster tables – segment mass and CoM weighting.
- Timepoints: how take-off and landing were identified.
- Use of absolute error/percentage error / RMSE (include equations)

Results

- 10 %

Tips:

- Do not interpret your results here
- Don't duplicate results
- Introduce figures or tables with written text
- Label figures/tables clearly (title, units, axes)

- Present the findings clearly and concisely
- Should include:
- Graphs/tables for:
 - Force-time and CoM displacement curves.
 - Jump height estimates from all methods for CMJ and SJ.
- Summary table comparing methods and the CoM (control) values.

Discussion & Conclusion

- 20 %

Tips:

- Make links between theory (intro) and results.
- Avoid repeating the results – focus on *interpretation*.

Discussion

- Analyse the results and draw conclusions.
- Compare each method to the control CoM method.
- Identify which method is most accurate and why.
- Interpret the performance differences between CMJ and SQJ.
- Compare your findings to existing literature (agreement or disagreement).
- Identify possible sources of error or limitations (e.g., marker placement, resampling issues).

Conclusion (max 200 words)

- Restate the aim and briefly summarise key findings.
- Highlight the best method and implications for practice or future work.

References



Tips:

- Use Endnote or other referencing software to make it easier to manage
- Do not use websites as sources of information

- Cite all papers, books, and online resources used.
- Alphabetical order
- Use Harvard referencing style

Additional tips

- Read around the topic
- Use more research papers than ones you have been provided with
- Use research papers as a guide for structuring your work, writing style etc.
- Leave yourself enough time to check through your work thoroughly
- Read your work out loud to help identify any errors that your brain might auto correct when you read in your head.
- Refer to learning materials provided to help you with your work
- Referencing info: [Referencing - Business & Management - LibGuides at Aston University](#)

Submission

- Due 10 th April 4pm
- Submit report as a pdf
- Submit your Matlab code and all data (excel files) as a zipped folder